HostingLint Milestone Roadmap

For: NLnet NGI Zero Commons Fund Application

Date: February 2026

Total Budget: EUR 35,000

Timeline: 12 months (Q2 2026 - Q1 2027)

Milestone Overview

Milestone	Description	Hours	Cost (EUR)	Timeline
M1	Rule Engine Expansion	120	6,000	Months 1-3
M2	Auto-Fix and Plugin System	140	7,000	Months 3-5
M3	Developer Experience	100	5,000	Months 5-7
M4	Documentation and Community	80	4,000	Months 7-9
M5	Security and Quality	80	4,000	Months 9-10
M6	v1.0 Release and Ecosystem	100	5,000	Months 10-12
PM	Project Management	80	4,000	Throughout
Total		**700**	**35,000**	

M1: Rule Engine Expansion (120 hours, EUR 6,000)

Objective: Expand from 28 to 50+ rules across all three platforms.

Deliverables

1. **8 Priority 1 rules** — High-impact security vulnerabilities
2. `security-php-deserialization` — Insecure unserialize() (CWE-502)
3. `security-php-ssrf` — Server-side request forgery (CWE-918)
4. `security-php-weak-crypto` — Weak password hashing (CWE-327)
5. `security-perl-sql-injection` — SQL injection via interpolation (CWE-89)
6. `security-perl-cmd-injection` — Command injection detection (CWE-78)
7. `openpanel-security-bind-mounts` — Dangerous host path mounts
8. `openpanel-security-secrets-in-env` — Secrets in ENV variables

9. `security-insecure-random` — Weak RNG for security purposes

1. **6 Priority 2 rules** — PHP 8.2/8.3 compatibility
2. `php-compat-dynamic-properties` — Deprecated in PHP 8.2
3. `php-compat-utf8-encode` — Removed in PHP 8.2
4. `php-compat-dollar-brace` — Deprecated in PHP 8.2
5. `php-compat-null-to-internal` — Deprecated in PHP 8.1
6. `php-compat-implicit-float-conversion` — PHP 8.1 change
7. `php-compat-serializable-interface` — Deprecated in PHP 8.1

1. **4 Priority 3 rules** — WHMCS 8.11-8.13 specific
2. `whmcs-payment-gateway-deprecated`
3. `whmcs-capsule-orm-migration`
4. `whmcs-hook-facade`
5. `whmcs-strict-types`

1. **4+ Priority 4-5 rules** — cPanel UAPI + OpenPanel container
2. `perl-cpanel-api1-functions`
3. `perl-cpanel-api2-to-uapi`
4. `openpanel-security-host-network`
5. `openpanel-security-latest-tag`

Success Criteria

- 50+ total rules implemented and tested
- 100% test coverage on all rule files
- < 200 ms analysis time for 1000 lines
- Vulnerable examples for all new rules

M2: Auto-Fix and Plugin System (140 hours, EUR 7,000)

Objective: Enable automated code fixes and community rule contributions.

Deliverables

1. **Auto-fix engine** — Machine-applicable fixes for 30+ rules
2. `FixAction` interface with range-based replacements
3. `--fix` CLI flag for automatic application
4. Interactive fix preview mode
5. Dry-run support

1. **Plugin system** — Load external rule packages

2. Plugin interface: `HostingLintPlugin { name, rules }`
3. npm package loading: `plugins: ['@hostinglint-rules/my-rules']`
4. Local file loading for development
5. Plugin validation and sandboxing

1. **Shared configurations** — Preset rule sets
2. `recommended` — balanced defaults
3. `strict` — all rules at error level
4. `security-only` — security rules only
5. Extendable presets via config

Success Criteria

- 30+ rules with working auto-fix
- Plugin API documented with example package
- 3 preset configurations published

M3: Developer Experience (100 hours, EUR 5,000)

Objective: Bring HostingLint into developer workflows.

Deliverables

1. **VS Code extension** — inline warnings, quick fixes
2. **PhpStorm plugin** — via Language Server Protocol
3. **Pre-commit hook** — `hostinglint check --staged`
4. **GitHub Action** — `hostinglint/action@v1`

Success Criteria

- VS Code extension published on marketplace
- Pre-commit hook documented and tested
- GitHub Action with SARIF upload

M4: Documentation and Community (80 hours, EUR 4,000)

Objective: Comprehensive documentation for users and contributors.

Deliverables

1. **Documentation website** — GitHub Pages, auto-generated rule catalog

2. **Rule documentation** — each rule with examples, fix suggestions, SARIF output
3. **Migration guides** — PHP 7 to 8, API1 to UAPI, cPanel to OpenPanel
4. **Video tutorials** — CLI walkthrough, CI/CD setup, rule development

Success Criteria

- Documentation website live at hostinglint.dev or GitHub Pages
- All 50+ rules documented with examples
- 3+ migration guides published

M5: Security and Quality (80 hours, EUR 4,000)

Objective: Security audit and quality assurance.

Deliverables

1. **Security audit** — via NLnet/Radically Open Security
2. **Accessibility review** — CLI output for screen readers
3. **Performance optimization** — benchmarking, profiling, caching
4. **Integration testing** — real-world WHMCS/cPanel/OpenPanel modules

Success Criteria

- Security audit completed with all findings addressed
- CLI output accessible (WCAG-compatible text)
- Performance under target with 50+ rules

M6: v1.0 Release and Ecosystem (100 hours, EUR 5,000)

Objective: Stable release and ecosystem integration.

Deliverables

1. **npm v1.0 release** — `@hostinglint/core` and `hostinglint` CLI
2. **ModuleForge integration** — updated @hostinglint/core dependency
3. **Community governance** — GOVERNANCE.md, maintainer guidelines
4. **Ecosystem integrations** — cPanel DevKit, WHMCS Marketplace mentions

Success Criteria

- v1.0.0 published on npm

- Stable API with semver guarantees
- Community governance model documented

Timeline Visualization

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2026
Q2 (Apr-Jun)    ═══════════ M1: Rule Expansion (50+ rules)
                     ═══════════════ M2: Auto-Fix + Plugins
Q3 (Jul-Sep)                 ══════════ M3: IDE Extensions
                                  ═══════ M4: Documentation
Q4 (Oct-Dec)                           ══════ M5: Security Audit
                                              ═══════ M6: v1.0 Release
2027
Q1 (Jan)        PM: Project Management (throughout) ──────────────────►
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Prepared for NLnet NGI Zero Commons Fund Application, February 2026